AN INTELLIGENT ARM FOR IVF

In in-vitro fertilisation, a woman's egg is fertilised in a lab with a sperm, and the resultant embryo transferred to her uterus. Now, AI is being used to sharpen the process

THE COSTS OF IVF WITH AI

1. BLOOD WORK & COUNSELLING
Bloods, egg, semen analysis; scans; treatment plan
Starts at: ₹10,000 at standard clinics/ ₹50,000 at AI-aided clinics*

2. OVARIAN STIMULATION
Injectable hormones for ovaries to mature eggs + scans
From ₹50,000/ ₹1 lakh for AI-aided

3. TRIGGER & EGG RETRIEVAL
Hormone injection for ovulation and ovum pick-up (OPU)
From ₹30,000/ ₹60,000 for AI-aided

4. FERTILISATION & EMBRYO CULTURE
Growing the embryo in a lab for 5-6 days (blastocyst)
From ₹30,000/ ₹60,000 for AI-aided

5. FRESH EMBRYO TRANSFER (ET)
Single or double embryo placed in uterus
From ₹15,000/ ₹5,000 for AI-aided

**Clinics do not charge extra for AI-aided IVF, but premium IVF chains use them, thus making costs higher*

AI ACROSS THE IVF PROCESS

1. PRE-IVF CONSULTATION
AI analyses medical history, and age and prior treatment data to estimate fertility chances and guide couples on whether IVF is the right next step

2. OVARIAN STIMULATION
Machine learning models predict how each patient will respond to hormone injections, helping doctors personalise drug doses



Photo courtesy BIRLA FERTILITY & IVF

3. EGG & SPERM ANALYSIS
Computer vision systems assess sperm motility, check egg quality with more precision than manual microscopy

4. FERTILISATION & CULTURE
AI-powered incubators monitor embryo development, detecting subtle changes invisible to the human eye; robotic systems ensure precise handling

5. EMBRYO SELECTION
Algorithms scan time-lapse images to rank embryos by their implantation potential, helping doctors choose the strongest embryo

6. IMPLANTATION
Predictive models estimate likelihood of embryo implantation. AI chatbots, virtual assistants offer reminders, emotional support

Yet money is only one barrier. AI in IVF has scientific and practical limitations. They rely heavily on data for accuracy and poor or biased datasets can distort predictions. Even when results are reliable, the interpretability gap remains: AI may suggest one embryo over another without saying why.

"We need more published data comparing AI-based outcomes with those from embryologists. Right now, there are some discrepancies," says Ghatak. He also points to the regulatory lag: privacy, transparency and data-sharing rules are evolving too slowly compared to the pace of technology. Integration is a challenge too—uniform training, standardised lab practices and trust-building all take time. As Ghatak notes, "High costs, patchy regulation and limited evidence are holding AI back."

FUTURE PROSPECTS

Despite the current drawbacks, the direction is clear. With stronger datasets, AI can move from a promising add-on to an integral part of IVF. For patients like Aisha, each advance brings the possibility of making treatment not only more precise but also more affordable and accessible.

Looking ahead, experts see AI moving beyond embryo selection into IVF counselling and personalised treatment. "AI may play a supporting role in counselling—providing credible information, reminders and guidance—but empathy and emotional support will always remain at the heart of IVF," says Benara.

Agarwal agrees. "IVF is not just a medical procedure, it is often the last hope for couples who have faced years of infertility struggles," she says. Global success rates hover at 40-50 per cent, meaning many couples endure multiple cycles. Studies show failed cycles often cause depression and anxiety for months. "Improving IVF success rates is about more than data. Even incremental gains spare families the pain of repeated loss."

From smarter embryo imaging to digital assistants that track well-being, AI will keep reshaping fertility care. While genetically engineered 'designer babies' are perhaps an impossible dream, hope has never looked so real outside of science fiction. ■